

OMAR ABDELHADI

 omarabdelhadi.com

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Education

Carleton University

Sep 2020 – Dec 2024

Bachelor of Computer Science (Honours) - Artificial Intelligence and Machine Learning, GPA: 3.65/4 Ottawa, Ontario

Technical Skills

Languages: Python, Java, C/C++/C#, **Tools & Frameworks:** Spring Boot, Node.js, React, Tailwind, Next.js, Docker, Jenkins, Git, GitHub, Bitbucket, Vite, NumPy, Pandas, Tanzu

Cloud & DevOps: AWS, Azure, CI/CD Pipelines, Kubernetes, Terraform, Jira, Blackduck

Databases: MySQL, MongoDB, Microsoft SQL Server

Operating Systems: Linux, Mac, Windows

Experience

Software Developer

Jan 2024 – Present

3 Canadian Space Division (DND)

Ottawa, Ontario

- Engineered and maintained **Spring Boot** microservices and automated **cron pipelines** that ingest and process multi-source satellite data daily, improving data reliability and accessibility for mission analysis.
- Developed full-stack features using **React**, **TypeScript**, and **Java**, building interactive dashboards, orbital maps, and analytical tools used by internal stakeholders to monitor key operational metrics.
- Managed deployment workflows across **Azure** and **Tanzu**, integrating CI/CD automation to reduce deployment friction and ensure consistent delivery across environments.
- Built and maintained comprehensive **JUnit** and **Vitest** test suites, increasing coverage, reducing regression issues, and improving long-term maintainability of both backend and frontend systems.

Software Engineering Intern

Sep 2022 – Aug 2023

Barracuda Networks LTD

Ottawa, Ontario

- Enhanced **.NET C#** microservices and REST APIs, delivering new features and resolving production issues for a large-scale RMM platform.
- Developed **React** components and integrated them with backend APIs to improve UI responsiveness and data flow.
- Leveraged **AWS** for service monitoring and implemented **CI/CD pipelines** to automate code scans, enforce quality gates, and manage package licensing.
- Built a low-level Windows system agent using **C#** and scripting to collect hardware, software, and performance telemetry for downstream analytics.

Web Developer Intern

May 2022 – Aug 2022

Getit Local Technologies

Ottawa, Ontario

- Developed **React Native** mobile features and optimized **React** web components across iOS, Android, and desktop.
- Designed and optimized **MongoDB** queries to handle large datasets efficiently.
- Implemented **Node.js REST APIs** to streamline backend-frontend communication.

Teaching Assistant

Jun 2021 – Dec 2024

Carleton University

Ottawa, Ontario

- Mentored students in programming labs for **Python**, **JavaScript**, and **Java**, guiding them through best practices in debugging, software design, and problem-solving.
- Led weekly technical workshops and office hours, breaking down complex concepts into clear explanations to improve comprehension and coding proficiency.
- Collaborated with professors to refine course materials, improve course structure, and enhance the overall curriculum.

Projects

Canny Edge Detector | Python

Jan 2024

- Implemented the **Canny Edge Detection** algorithm, utilizing techniques such as **Gaussian blurring**, gradient calculation, and edge tracking by hysteresis to detect edges in images.
- Visualized edge detection results on various images with varying **sigma** and **threshold** values, showcasing potential applications in lane detection technology.

AED Simulation Software | QT C++

Dec 2023

- Developed a prototype for **AED Simulation** to replicate key functions of an Automated External Defibrillator (AED), enhancing user preparedness for cardiac emergencies.
- Designed the system to guide users through heart rhythm analysis, shock administration, and provide real-time assistance during emergencies.